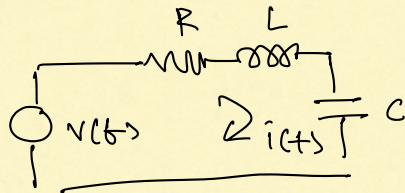


Lecture-13

Circuit:



Question: Find the solution to the steady state current $i(t)$ if $v(t)$ is a T -periodic signal.

$$\frac{d^2 i}{dt^2} + \frac{R}{L} \frac{di}{dt} + \frac{1}{LC} i = \frac{1}{L} v'(t).$$

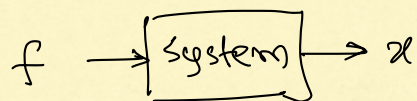
Call $\frac{1}{L} v'(t) = f(t)$ and $i(t) = x(t)$.

$$\frac{R}{L} = 2k \quad , \quad k > 0$$

$$\frac{1}{LC} = b \quad , \quad b > 0$$

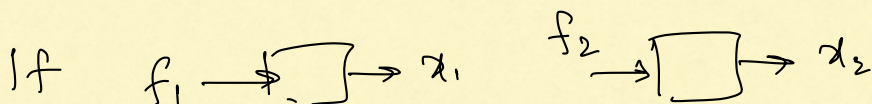
We have: $x'' + 2kx' + bx = f$

It is useful to think of f as input to a system and x as the output of the system.



Main Observations:

(1) This system is linear.



then $\alpha_1 f_1 + \alpha_2 f_2 \rightarrow \alpha_1 x_1 + \alpha_2 x_2$

Why is this true?

We are given: $x_1'' + 2kx_1' + bx_1 = f_1$

$$x_2'' + 2kx_2' + bx_2 = f_2$$

Now, if you consider $x = \alpha_1 x_1 + \alpha_2 x_2$

where α_1, α_2 are scalars:

$$x'' + 2kx' + bx = (\alpha_1 x_1 + \alpha_2 x_2)'' + 2k(\alpha_1 x_1 + \alpha_2 x_2)' + b(\alpha_1 x_1 + \alpha_2 x_2)$$

$$= \alpha_1 x_1'' + \alpha_2 x_2'' + 2k\alpha_1 x_1' + 2k\alpha_2 x_2' + b\alpha_1 x_1 + b\alpha_2 x_2$$

$$= \alpha_1 (x_1'' + 2kx_1' + bx_1)$$

$$+ \alpha_2 (x_2'' + 2kx_2' + bx_2)$$

$$= \alpha_1 f_1 + \alpha_2 f_2.$$

② If $f = e^{jn\omega t}$ then $x = z_n \cdot e^{jn\omega t}$

where z_n is a constant.

We can find z_n :

$$x'' + 2kx' + bx = f$$

Choose $f = e^{jn\omega t}$ and set $x = z_n e^{jn\omega t}$.

$$x' = z_n \cdot jn\omega \cdot e^{jn\omega t}$$

$$x'' = z_n (-1) \cdot n^2 \omega^2 \cdot e^{jn\omega t}$$

$$\therefore -z_n n^2 \omega^2 \cdot e^{jn\omega t} + j 2kn\omega z_n \cdot e^{jn\omega t} + b \cdot z_n \cdot e^{jn\omega t} = e^{jn\omega t}$$

$$\therefore Z_n [-n^2 \omega^2 + j2kn\omega + b] = 1$$

$$\therefore Z_n = \frac{1}{(b - n^2 \omega^2) + j(2kn\omega)}$$

$$e^{jn\omega t} \rightarrow \boxed{\text{System}} \rightarrow Z_n e^{jn\omega t}$$

$\downarrow$ $\downarrow$
 If $f = \frac{v'(t)}{L}$ then we have $i = x(t)$

Suppose we feed a linear combination

$$\sum_{n=-N}^{+N} c_n e^{jn\omega t} \rightarrow \boxed{\text{Sys.}} \rightarrow \sum_{n=-N}^{+N} c_n Z_n e^{jn\omega t}$$

$\downarrow$
 If this is an approximation
 to $\frac{v'(t)}{L}$

$\downarrow$
 we hope that this
 is an approximation
 to $i(t)$.

Choose c_n to be the

Fourier series coeff of $\frac{v'(t)}{L}$, i.e. $\frac{v'}{L} \xleftrightarrow{\text{F.S.}} c_n$

If $v(t) \xleftrightarrow{\text{F.S.}} b_n$ and $v'(t)$ exists for all t

$$\text{then } \frac{v'(t)}{L} \xleftrightarrow{\text{F.S.}} \frac{b_n \cdot jn\omega}{L}$$

$$\text{Thus, we have } c_n = \frac{jn\omega b_n}{L}$$

∴ The approximation to the current is $\sum_{n=-N}^{+N} c_n \cdot Z_n e^{jn\omega t}$

$$= \sum_{n=-N}^{+N} \frac{jn\omega / L}{(b - n^2\omega^2) + j(2kn\omega)} \cdot b_n \cdot e^{jn\omega t}$$

Theorem: (based on Chap 23 of Körner "Fourier Analysis")

If $V(t) \xleftrightarrow{\text{F.S.}} b_n$ and if $V'(t)$ is continuous at all t and further if $\sum_{-\infty}^{+\infty} |nb_n| < \infty$ then

$$\sum_{n=-N}^{+N} \frac{jn\omega / L}{(b - n^2\omega^2) + j2kn\omega} b_n \cdot e^{jn\omega t} \rightarrow i(t)$$

in the uniform sense. → Call this H_n

Compare the Partial sums of Fourier series of $V(t)$ and $i(t)$:

$$\sum_{n=-N}^{+N} b_n \cdot e^{jn\omega t} \quad \text{and} \quad \sum_{n=-N}^{+N} H_n \cdot b_n \cdot e^{jn\omega t}$$

↓ ↓

Partial sum for $V(t)$ Partial sum for $i(t)$.

For each n :

$$\begin{matrix} n^{\text{th}} \text{ Fourier Series} \\ \text{coeff. of } i(t) \end{matrix} = H_n \times \begin{matrix} n^{\text{th}} \text{ Fourier Series} \\ \text{coeff. of } V(t) \end{matrix}$$

H_n = gain experienced by the n^{th} harmonic component of $V(t)$